

Document 08 — Frontend Architecture & User Interface Blueprint

Product Name: HQ

Tagline: Your AI Headquarters

Version: 1.0

Purpose

This document defines the architecture of the HQ frontend.

It establishes how the application is organized, how users navigate through the platform, how components communicate, and how the user experience remains consistent across the web and future mobile applications.

The frontend is designed to be modular, scalable, responsive, accessible, and AI-native.

Frontend Philosophy

HQ is not a collection of pages.

HQ is a living Headquarters.

Every interaction should make users feel like they are walking through an intelligent executive workspace where AI teams collaborate in real time.

Technology Stack

Framework

- Next.js (App Router)
- React
- TypeScript

Styling

- Tailwind CSS
- CSS Variables
- Design Tokens

UI Components

A reusable component library shared across Web and Mobile.

Examples:

- Buttons
- Cards
- Dialogs
- Inputs
- Dropdowns
- Tabs
- Tables
- Timeline
- AI Chat
- Executive Cards

State Management

Separate state into three categories:

Server State

Managed using TanStack Query.

Examples:

- User profile
- Missions
- Dashboard
- Analytics

Client State

Managed using Zustand.

Examples:

- Sidebar
- Theme
- Current Executive
- Active Workspace

Form State

Managed using React Hook Form.

Examples:

- Login
- Mission Creation
- Brand Setup
- Settings

Route Architecture

/

└─ Landing

└─ Pricing

└─ Features

└─ Documentation

└─ Blog

|

└─ Auth

| └─ Login

| └─ Register

| └─ Forgot Password

|

└─ App

└─ Dashboard

└─ Headquarters

└─ Executive Board

└─ Mission Control

└─ Assets

└─ Analytics

└─ Billing

└─ Settings

└─ Profile

Layout Architecture

The application uses a persistent layout.

Includes:

- Left Sidebar
- Top Navigation
- Main Workspace
- Right Activity Panel

The layout never reloads while navigating.

Dashboard Structure

Displays:

- CEO Welcome
- Mission Overview
- Executive Board
- Active Missions
- Today's Progress
- Notifications
- AI Insights
- Subscription Status

Component Architecture

Page



Section



Feature Component



Reusable Component



Primitive UI Component

Every component should have a single responsibility.

Executive Board UI

Each executive card displays:

- Avatar
- Name
- Department
- Status
- Current Task
- Progress
- AI Activity Indicator

Clicking a card opens a detailed executive workspace.

Mission Control UI

Mission Control is the centerpiece of HQ.

Users can:

- Create missions
- Monitor progress
- View task assignments
- Observe executive collaboration
- Approve outputs
- Review deliverables

AI Collaboration Timeline

A live timeline displays:

- CEO Analysis
- Chief of Staff Planning
- Department Assignment
- Executive Activity

- Quality Review
- Final Approval

Users should feel like they are watching a real executive team at work.

AI Streaming Experience

AI responses stream progressively.

Executives update independently.

The interface always shows:

- Thinking
- Researching
- Designing
- Writing
- Reviewing
- Completed

This provides transparency and keeps users engaged.

Navigation Principles

Primary Navigation

- Dashboard
- Mission Control
- Executive Board
- Assets
- Analytics

Secondary Navigation

- Billing
- Settings
- Help
- Profile

Responsive Design

Desktop

Primary experience.

Tablet

Adaptive multi-column layout.

Mobile

Single-column layout with optimized navigation.

The mobile application will share the same design language.

Accessibility

HQ follows accessibility best practices:

- Keyboard navigation
- Screen reader support
- High contrast support
- Focus indicators
- Semantic HTML
- Accessible forms

Loading Experience

Replace generic loaders with executive activity.

Examples:

- CEO reviewing...

- Research gathering insights...
- Design creating assets...
- Marketing preparing strategy...

The system should always communicate meaningful progress.

Error Experience

Errors should be clear and actionable.

Instead of:

"Something went wrong."

Display:

"The Design Director couldn't complete this task. Please retry or assign it again."

Empty States

Every empty state should encourage action.

Example:

"No missions yet."

"Launch your first mission and let your Executive Board get to work."

Animation Principles

Animations must:

- Communicate progress
- Reinforce hierarchy
- Feel smooth
- Never distract

Micro-interactions should support usability rather than decorations

Theme System

Supports:

Light Mode

Dark Mode

Uses design tokens so branding can evolve without rewriting components.

Frontend Folder Structure

apps/web

|

└─ app/

└─ components/

| └─ ui/

| └─ executives/

- | | └─ missions/
- | | └─ dashboard/
- | | └─ analytics/
- | └─ shared/
- |
- | └─ features/
- | └─ hooks/
- | └─ lib/
- | └─ services/
- | └─ store/
- | └─ styles/
- | └─ types/
- └─ utils/

Frontend Design Principles

Every screen must be:

Executive

Professional

Fast

Modular

Accessible

Consistent

Responsive

AI-first

Success Criteria

The frontend succeeds when users:

Understand where they are.

Understand what the Executive Board is doing.

Feel in control of every mission.

Trust the AI process.

Enjoy using HQ every day.

Conclusion

The HQ frontend is designed to deliver an immersive executive experience rather than a traditional dashboard.

Every component, animation, and interaction should reinforce the idea that users are operating their own AI Headquarters powered by a collaborative executive organization.